Alcohol Dose and Mental Arithmetic Performance

A Randomized Block Experiment on The Islands

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1 Introduction

Alcohol consumption is a common social behavior, especially in the college-age demographic, and it has the potential to influence cognitive functions such as memory, attention, and problem solving. Because mental arithmetic underlies many everyday and academic tasks, it is worth asking how alcohol affects it and whether the **amount** consumed matters. Our investigative questions are: (1) does drinking alcohol affect performance on a timed mental-arithmetic task, and (2) does the size of that change depend on the dose and on the difficulty of the task?

Going in, we expected alcohol to reduce performance, and we expected the reduction to grow with dose. We also suspected the effect would be larger on harder problems, which demand more working memory and controlled attention, than on easy, well-practiced problems. We tested these expectations with a designed experiment on the virtual population of **The Islands** (Providence Island), where each inhabitant can be sampled, given a measured dose of alcohol, and timed on a standardized arithmetic task.

2 Design of the Experiment

2.1 From the original design to a corrected one

Our first design gave **every subject all six** alcohol-by-difficulty conditions in a single session, treating each subject as a block. This is not a valid randomized block design for alcohol: doses were administered in sequence with no washout, so cumulative dose, fatigue, and practice on the task are confounded with the alcohol level, and the handful of subjects could not support generalization. We therefore redesigned the study as a **between-subjects** experiment: each islander is measured once at a sober **baseline**, then randomly assigned to exactly **one** treatment and re-tested. Random assignment is what licenses the causal comparison: any systematic difference between dose groups is attributable to dose, not to who landed in which group.

2.2 Response: change from baseline (a proxy for math skill)

Islanders differ enormously in baseline arithmetic ability, and that ability is not the thing we want to measure. Rather than analyze raw post-treatment scores, we use each subject's own sober baseline as a proxy for their underlying math skill and analyze the **change**, $D = \text{baseline} - \text{post}$: the number of correct answers *lost* relative to that subject's own sober performance at the same difficulty. Differencing against the personal baseline removes between-subject ability so the analysis isolates the alcohol-induced change.

2.3 Treatments, blocks, and randomization

This is a **randomized block design** with a 2×3 treatment layout, analyzed with replication (a generalized RBD):

- **Treatment factor — alcohol dose:** 1, 2, or 3 shots of vodka (30 mL each). The sober baseline supplies the no-alcohol reference, so no separate 0-shot group is needed.
- **Blocking factor — task difficulty:** Easy or Hard. Difficulty is the largest known source of variation in performance change, so we block on it to estimate the dose effect more precisely; replication within blocks also lets us test whether the dose effect differs by difficulty.
- **Experimental units:** islanders aged 21+, randomly sampled from households on Providence Island and randomly assigned to a dose within each difficulty block. Each unit receives one treatment, eliminating the carryover that invalidated the original design.

2.4 Sample size and power

We fixed the sample size in advance with a power analysis. Treating dose as a three-level factor, detecting a **large** effect (Cohen's $f = 0.40$; Cohen, 1988) at $\alpha = 0.05$ with power 0.80 requires 22 islanders per dose level:

```
pwr.anova.test(k = 3, f = 0.40, sig.level = 0.05, power = 0.80)$n
```

```
## [1] 21.10364
```

We collected **60 islanders**, balanced as **10 per cell** (20 per dose, 30 per difficulty block), giving power 0.78 for a large dose effect and effectively full power for the large difficulty effect. The design is completely balanced, so all factor effects are orthogonal. As we report below, the realized dose effect was smaller than the large one we planned for, a point we return to in the Discussion.

3 Results and Interpretation

Table 1 and Figure 1 summarize the mean drop D in each cell. The pattern is striking: on Easy tasks the drop hovers near one point regardless of dose, whereas on Hard tasks it climbs steadily from 2.8 at one shot to 6.4 at three.

Table 1: Mean drop in score (sober baseline minus post-treatment) by difficulty block and dose. Each cell is 10 islanders.

	1 shot	2 shots	3 shots
Easy	1.2	1.0	1.3
Hard	2.8	4.2	6.4

3.1 Primary analysis: randomized block ANOVA

```
fit <- aov(diff ~ difficulty * dose, data = d)   # block * treatment, change score
```

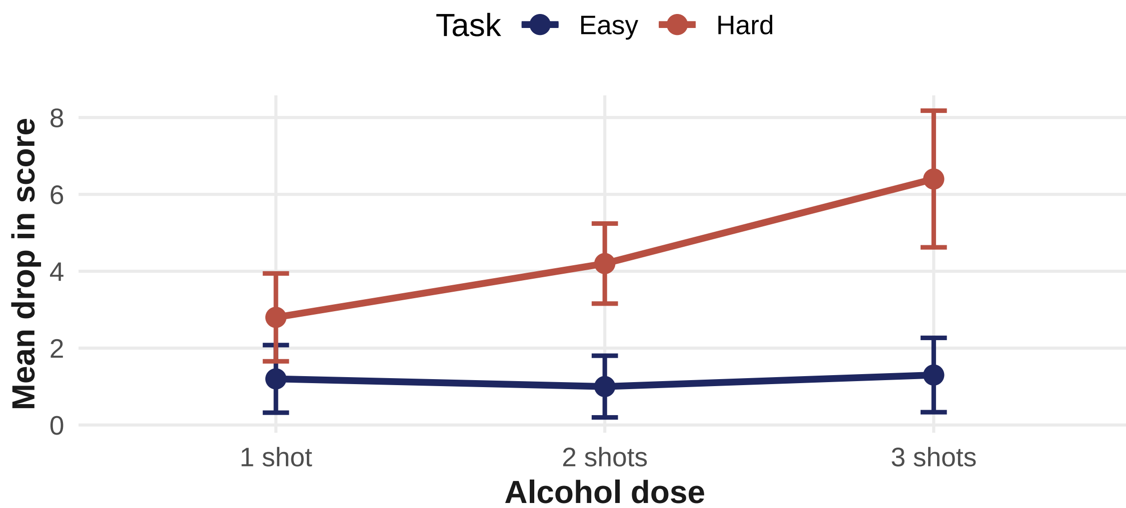

Figure 1: Mean drop in arithmetic score (with one-standard-error bars) by alcohol dose, for Easy and Hard tasks. Alcohol degrades hard-task performance in a dose-dependent way; easy-task performance is essentially unaffected.

Table 2: Change-score ANOVA. Difficulty is the blocking factor, dose the treatment.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
difficulty	1	163.35	163.35	12.40	0.0009
dose	2	35.63	17.82	1.35	0.2672
difficulty:dose	2	30.70	15.35	1.17	0.3195
Residuals	54	711.30	13.17	NA	

The blocking factor, **difficulty**, is highly significant ($F_{1,54} = 12.4$, $p = 8.8 \times 10^{-4}$): hard tasks lose far more ground under alcohol than easy ones, which confirms that blocking on difficulty was the right choice. The **dose** main effect averaged over both blocks is not significant ($F_{2,54} = 1.35$, $p = 0.27$), and neither is the block-by-treatment interaction ($p = 0.32$) as an omnibus test: both are diluted by the flat Easy block.

3.2 The dose effect is carried by hard tasks

Because dose is ordered, a planned linear trend test is the more powerful and direct way to ask "does the amount matter."

```
fit_lin <- lm(diff ~ difficulty * shots_n, data = d)  # dose as a numeric trend
```

The trend model explains 24% of the variance in D ($p = 0.0013$). Within Easy tasks the dose slope is essentially zero (+0.05 points/shot); within Hard tasks each additional shot costs **1.80 correct answers** (SE 0.95, two-sided $p = 0.067$; one-sided $p = 0.034$ given our directional prediction).

3.3 Robustness: analysis of covariance

The change score implicitly assumes a one-point gain in sober baseline maps to a one-point gain in treated score. We relax that by regressing the raw post-treatment score on the baseline, which also corrects for regression to the mean:

```
anc <- lm(response ~ baseline + difficulty * dose, data = d)  # ANCOVA
```

Table 3: ANCOVA (Type II): post-treatment score adjusted for the sober baseline.

	Sum Sq	Df	F value	Pr(>F)
baseline	3641.74	1	286.47	0.0000
difficulty	74.47	1	5.86	0.0190
dose	37.88	2	1.49	0.2346
difficulty:dose	47.63	2	1.87	0.1636
Residuals	673.76	53	NA	

The fitted baseline coefficient is 1.11, close to the 1.0 the change score imposes, so the two models agree and the residual scatter barely moves (3.57 vs. 3.63). Baseline ability and difficulty dominate, and the omnibus dose effect stays non-significant ($p = 0.23$). The one shift is that the planned hard-task dose contrast sharpens to -2.07 correct per shot (two-sided $p = 0.038$, one-sided 0.019), and the difficulty-by-dose interaction moves to $p = 0.058$. Because the baseline slope varies modestly across doses (equal-slopes test $p = 0.08$), we report the ANCOVA **alongside**, not in place of, the change-score model. Both lead to the same conclusion.

3.4 Assumptions

Residuals are consistent with normality (Shapiro-Wilk $p = 0.77$) and with equal variances across cells (Levene $p = 0.13$), so the ANOVA is well calibrated (diagnostic plots in the Appendix).

4 Discussion

Summary. Alcohol's effect on mental arithmetic depends on how hard the task is. Easy, over-learned problems sit near the 40-question ceiling and are essentially unaffected by one to three shots. Hard problems degrade in a dose-dependent way, losing roughly 1.8 correct answers for every additional shot (significant under ANCOVA, marginal under the change score). Task difficulty was the dominant, clearly significant factor; the dose effect is real but concentrated in the hard block.

Does this make sense? Yes. Simple arithmetic is automatic and robust, while hard arithmetic leans on working memory and controlled attention — exactly the faculties that alcohol is known to impair first. The widening Easy-Hard gap with dose is consistent with the *alcohol-myopia* account (Steele & Josephs, 1990), in which alcohol narrows the scope of attention and so disrupts effortful, attention-demanding cognition more than automatic processing. The flat Easy curve is a textbook ceiling effect: there is almost no room to fall when sober performance is already near maximum.

Limitations.

1. *Simulated population.* The Islanders are a virtual population, so the effects reflect the simulation's model and external validity to real drinkers is limited; the sample is also drawn from a single island.

2. *Practice (warm-up) confound.* Every subject took the sober baseline before the treated test, so practice warms them up and inflates the post-treatment score, which shrinks D. This biases the estimated *absolute* impairment toward zero. Because dose was randomized, the practice gain is balanced across dose groups and largely cancels in the dose-to-dose comparison, but a sober test–retest control arm would be needed to net it out cleanly.
3. *Dose is not exposure.* Shots were not titrated to body water (weight, sex) or tolerance, so a fixed dose produces different blood-alcohol levels and different impairment across people. This heterogeneity inflates within-cell variance and is part of why the study is underpowered.
4. *Underpowered for the realized effect.* We powered for a large effect ($f = 0.40$); the observed dose effect was smaller ($f \approx 0.22$), so detecting it at 0.80 power would need roughly 66 islanders per dose (about 198 total).
5. *Ceiling on easy tasks.* Sober easy scores sit at 38–40 of 40, leaving almost no room to fall, so a genuine easy-task effect could be masked.

Natural next steps follow directly: add a sober test–retest control arm to remove practice, record weight and sex to model blood-alcohol concentration, concentrate sampling on hard tasks, and extend the dose range beyond three shots.

5 References

Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.

Steele, C. M., & Josephs, R. A. (1990). Alcohol myopia: Its prized and dangerous effects. *American Psychologist, 45*(8), 921–933.

A Appendix

```r
TukeyHSD(fit, "dose")$dose   # pairwise dose comparisons (change score)
```

```
##     diff        lwr      upr     p adj
## 2-1 0.60 -2.1659458 3.365946 0.8605602
## 3-1 1.85 -0.9159458 4.615946 0.2494043
## 3-2 1.25 -1.5159458 4.015946 0.5248722
```

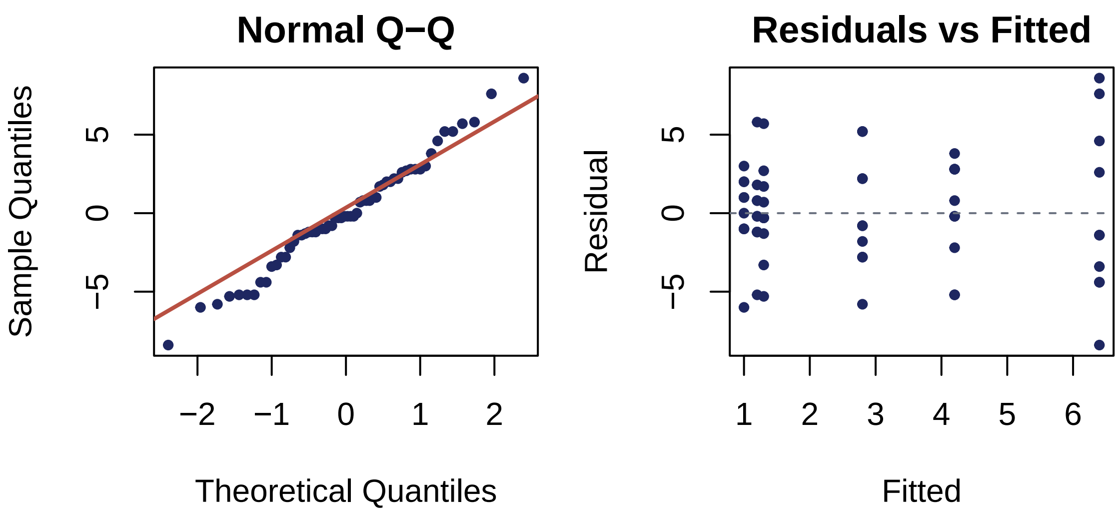

Figure 2: Diagnostic plots for the change-score model: residual Q-Q and residuals vs. fitted.